

LifeLink

Emergency Blood & Medicine Request Network

Adaptive Software Engineering Project

Roll Numbers: 2420030262-J.Nandini
2420030266-S.Sai Saranya
2420030364-B.Sai Charan

Guide Name: G.Lavanya Mam

PROBLEM STATEMENT

During medical emergencies, patients and their families may struggle to quickly find compatible blood donors, required medicines, and reliable support.

Current methods often depend on phone calls, personal contacts, social-media posts, or visiting multiple hospitals and pharmacies.

These methods can be unorganized, time-consuming, and may not provide clear information about donor availability, request status, or nearby medicine providers.

LifeLink addresses this problem through a centralized digital platform for discovering and coordinating emergency blood and medicine resources.

OBJECTIVES

Develop a centralized platform for emergency blood and medicine resource discovery.

Allow users to register, log in, manage profiles, and participate as donors or volunteers.

Enable donor search using blood group, location, and availability.

Allow users to create and track emergency blood requests.

Provide medicine search and pharmacy information in one system.

Use location services to help identify nearby resources.

Apply Agile Scrum and Adaptive Software Engineering through iterative development and feedback.

EXISTING SYSTEM

Current Approach: People search for blood donors through friends, hospitals, blood banks, phone calls, and social-media groups.

Medicine Search: Users may contact multiple pharmacies individually to check availability.

Coordination: Emergency requests are often handled manually without a single platform for tracking progress.

Limitations: Slow communication, scattered information, duplicate efforts, uncertain availability, and difficulty finding nearby resources quickly.

Technical Limitation: Manual approaches do not provide a unified workflow, centralized data management, or easy request-status tracking.

PROPOSED SYSTEM

LifeLink Platform: A centralized web application connecting users, blood donors, volunteers, and pharmacies.

Blood Support: Register donors, maintain availability, and search by blood group and location.

Emergency Requests: Create requests with blood group, units, hospital/location, urgency, and status.

Medicine Finder: Search medicines and view relevant pharmacy information.

Location Support: Integrate maps to help users discover nearby donors and pharmacies.

Adaptive Development: Use feedback after each sprint to modify and improve features.

Benefits: Faster resource discovery, better coordination, centralized information, and clearer request tracking.

TECHNOLOGY USED

Frontend: HTML5 • CSS3 • JavaScript • JSP

Backend: Java • Servlets • JDBC

Database: MySQL

Web/Application Server: Apache Tomcat

APIs: REST APIs • Google Maps API (planned advanced feature)

Project Management: Agile Scrum • Jira

Version Control: Git • GitHub

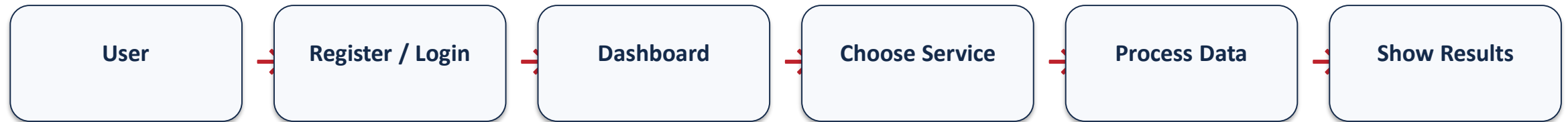
UI/UX Design: Figma

Testing: JUnit • Postman

Containerization: Docker • Docker Compose

AI Feature (advanced/planned): AI-assisted donor recommendation / Google Gemini API

WORKFLOW



Blood Flow: Search Blood → Enter Blood Group + Location → Database Search → Display Available Donors

Development Flow: Jira Backlog → Sprint → Development → Git/GitHub → Testing → Docker → Review & Feedback → Next Sprint

MODULES

1. User Authentication & Profile: Registration, login, logout, validation, and profile management.
2. Blood Donor Management: Donor registration, blood group, location, availability, and donor search.
3. Emergency Request Management: Create requests and track Open → Accepted → In Progress → Completed.
4. Volunteer Management: Volunteer registration, availability, and support for active requests.
5. Medicine Finder & Pharmacy: Search medicines and display pharmacy details and availability.
6. Location Services: Nearby donor/pharmacy discovery using map integration.
7. AI-Assisted Matching: Optional advanced feature for ranking suitable donors.
8. Administration & Notifications: Advanced/future management and alert features.

IMPLEMENTATION

Project Planning: Project concept, title, modules, Agile/ASE workflow, and technology stack have been defined.

Jira & Agile Plan: Epics, sprints, and task structure have been planned for iterative development.

Database Setup: MySQL installation has been verified. The LifeLink database and initial users table are the first database implementation steps.

Development Setup: Java/JSP, JDBC, MySQL, Apache Tomcat, and Docker have been selected for implementation.

Next Steps: Create Maven web project → configure Tomcat → connect JDBC → build registration → login → dashboard → donor module.

CONCLUSION & FUTURE SCOPE

Current Status: The project architecture, modules, technology stack, Agile/Jira workflow, and initial MySQL environment have been established.

Conclusion: LifeLink proposes a centralized emergency-support platform for blood donor discovery, emergency requests, medicine search, volunteer coordination, and location-aware assistance.

Future Scope: Real-time notifications, hospital/blood-bank integration, verified pharmacy data, stronger authentication, mobile application, advanced analytics, and improved AI-assisted matching.

Deployment Scope: Complete Docker and Docker Compose setup, automated testing, GitHub Actions CI/CD, and optional cloud/Kubernetes deployment if required.

Adaptive Engineering Value: Feedback from every sprint can be converted into new or modified backlog items, allowing the system to evolve.

THANK YOU